

Closing the Lagrange–Rvachev Loop

Exact cardinal synthesis by shifted up-functions, defect-completed factorizations, convergence dichotomies, and Lambert-scale conditioning

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Abstract

Let $u = \text{up}$ denote Rvachev’s compactly supported C^∞ function, normalized by $u(x) = F(1 - |x|)$ on $[-1, 1]$, where F is the Fabius function. For arbitrary distinct interpolation nodes $-1 \leq x_0, \dots, x_n \leq 1$, this report obtains an exact representation of every Lagrange cardinal polynomial $\ell_{n,j}$ on $[-1, 1]$ as a linear combination of only $n + 2$ shifted copies of one common dilate of u :

$$\ell_{n,j}(x) = \frac{1}{2^n} \sum_{r=0}^{n+1} b_{j,r} u\left(\frac{x - c_{n,r}}{2^{n+1}}\right), \quad c_{n,r} = 2^{n+1} - 2n - 3 + 2r.$$

The coefficients are given by a finite Bernoulli–Bell differential transform followed by a unit-triangular deconvolution whose symbol is the q -integer product $\prod_{m=1}^n [2^m]_x$. Thus the formula is exact, algorithmic, rational for rational nodes, and directly connected with the Thue–Morse factorization of that product.

Substitution into a Lagrange representation of u closes the requested loop. The apparent $(n + 1)(n + 2)$ double sum collapses to the same $n + 2$ up-atoms. In matrix form, the cardinal synthesis matrix is a right inverse of the atom-sampling matrix. Appending the restricted Thue–Morse defect functional makes the matrix square and invertible. The resulting coefficient loop is a rank-one projector; its null vector is represented by

$$D_n - I_n D_n = \omega_n + (E_n - I_n E_n),$$

where $D_n = t^{n+1} + E_n$ is the canonical polynomial-defect generator and ω_n is the nodal polynomial. This identifies the unique “ghost mode” missed by interpolation.

A crucial distinction is established. At every finite degree the loop is an exact identity, but the equispaced dyadic global interpolants occurring in the repository exhibit Runge instability and do not thereby become a convergent representation of u . Chebyshev–Lobatto nodes instead yield a uniformly convergent, superalgebraic loop and, by telescoping nested degrees, a genuine convergent series of shifted up-functions. Nevertheless, the endpoint dictionary is intrinsically coefficient-ill-conditioned. If $a_{n,j,r} = 2^{-n} b_{j,r}$ are the raw amplitudes, then

$$\sum_r |a_{n,j,r}| \geq \frac{\|\ell_{n,j}\|_\infty}{F((n+2)2^{-n})},$$

so the dyadic endpoint law implies

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \log \sum_r |a_{n,j,r}| \geq \frac{\log 2}{2}.$$

Thus stable function approximation can coexist with cancellations of size $\exp((\log 2/2 + o(1))n^2)$. Exact computations through degree 13 support a sharper Lambert–phase asymptotic conjecture. The report includes complete proofs of the new deductions, explicit low-degree examples, numerical experiments, conjectures, and a reproducible Python implementation.

Status convention. Statements labelled theorem, proposition, lemma, or corollary are proved in this report, possibly using a clearly identified theorem already established in the repository. Statements labelled conjecture or question are not claimed as proved. Numerical figures are diagnostics, not proof.

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1 Executive summary and principal results

The requested construction sits at the intersection of two exact but very different coordinate systems. The first is polynomial interpolation:

$$I_X f(x) = \sum_{j=0}^n f(x_j) \ell_{n,j}(x), \quad \ell_{n,j}(x) = \prod_{i \neq j} \frac{x - x_i}{x_j - x_i}.$$

The second is the finite common-scale dictionary of translates of Rvachev’s $u = \text{up}$. The repository proves that, at the sharp dyadic ratio 2^n , a specific right-endpoint block of $n + 2$ translates is a basis for the degree n polynomial subspace on a unit interval. The present report specializes that result to the entire Lagrange cardinal basis, writes the coefficients in closed finite form, composes the two maps, and studies the resulting loop.

The main conclusions are as follows.

- R1. Exact cardinal synthesis.** For arbitrary distinct nodes in $[-1, 1]$, every cardinal polynomial is represented by the same $n + 2$ shifted copies of u at common scale 2^{n+1} . Only the amplitudes depend on the node and cardinal index.
- R2. Bernoulli–Bell– q formula.** The amplitudes are explicit. Bernoulli numbers determine the logarithm of a finite differential multiplier; complete exponential Bell polynomials expand it; a Toeplitz inverse of the q -integer product $A_n(z) = \prod_{m=1}^n [2^m]_z$ completes the calculation.
- R3. Finite closure with dictionary collapse.** Substitution into $I_X u$ produces a double sum, but all cardinal polynomials use the same atoms. Grouping by atom leaves only $n + 2$ terms. Directly synthesizing the single interpolating polynomial gives exactly the same coefficients.
- R4. Defect-completed inverse and ghost mode.** The $(n + 1) \times (n + 2)$ atom-sampling matrix has a one-dimensional kernel. Adding the Thue–Morse polynomiality row gives an invertible square matrix. The associated coefficient projector differs from the identity by one explicit rank-one defect projector. The corresponding function is the canonical defect minus its Lagrange interpolant.
- R5. Stability is inherited from interpolation, not repaired by synthesis.** On $[-1, 1]$, the closed-loop function is exactly $I_X f$. Hence equispaced Runge instability survives unchanged. Chebyshev–Lobatto nodes yield superalgebraic convergence for u , but no geometric convergence because u is nowhere analytic.
- R6. An unavoidable Lambert-scale cancellation barrier.** All endpoint atoms are sampled exponentially close to the flat edge of u . The resulting coefficient ℓ^1 norm is at least the reciprocal of $F((n + 2)2^{-n})$. The known dyadic endpoint law forces quadratic-exponential growth in n .
- R7. A genuine self-series.** Telescoping nested Chebyshev–Lobatto interpolants gives an absolutely convergent series in $C[-1, 1]$, and each polynomial block has an exact finite up-synthesis. This closes the loop as an actual convergent series, not only as a sequence of finite identities.

The most important conceptual point is that there are two distinct notions of stability. The function-space operator can be stable or convergent while the internal amplitude vector is catastrophically large. The loop is therefore mathematically exact but, in the endpoint basis, numerically delicate.

2 Corpus audit, dependency boundary, and non-duplication

2.1 How the repository was treated

The repository is a living research corpus. Its current organization explicitly consolidates earlier stand-alone drafts into thematic volumes and records the absorbed sources and hashes. To avoid treating historical copies as independent results, the present audit uses the following hierarchy:

1. the primary exposition `Fabius_Function_and_Rvachev_Up.tex`;
2. the current canonical and consolidated frontier volumes;
3. their provenance tables and manifests for absorbed drafts;
4. archival paper transcriptions as bibliographic and historical sources.

Targeted full-text inspection focused on the terms *Lagrange*, *interpolation*, *polynomial synthesis*, *endpoint basis*, *Thue–Morse certificate*, *canonical defect*, *q-binomial*, *Bell*, *Bernoulli*, and *Lambert*. This is the appropriate non-duplication boundary because the repository itself designates the consolidated volumes as the current record and stores absorbed drafts only as provenance.

2.2 Exact ingredients imported from the corpus

Table 1: Dependency map. “Imported” means used as an established corpus result; all subsequent deductions are proved here.

Source	Imported result	Use in this report
Primary exposition	Normalization $u(x) = F(1 - x)$, sinc-product Fourier transform, probabilistic model, moments and cumulants, q -geometric Lagrange weights, and dyadic endpoint asymptotics	Basic notation, Bell coefficients, q specialization, and coefficient lower bound
<i>Up Polynomial Synthesis</i>	Sharp common-scale ratio 2^n ; the $n+2$ -atom endpoint basis; the multiplier H_n ; triangular A_n -deconvolution; Thue–Morse certificate; canonical defect	The exact synthesis engine specialized to cardinal polynomials
<i>Dyadic Comb Frontiers</i>	Global equispaced interpolation formulas, symmetry reductions, computed Runge growth, and the warning that finite interpolants need not converge	The unstable branch of the loop analysis
<i>Representation Frontiers</i>	Smooth Chebyshev representation and superalgebraic coefficient decay	Stable node design and the convergent hierarchical loop

2.3 What appears to be new relative to the audited consolidated corpus

No matching statement was found for the following combined results:

1. the explicit arbitrary-node Lagrange-cardinal specialization of the endpoint basis, including the Bell–elementary-symmetric formula;
2. the common-dictionary collapse of the substituted Lagrange/up double sum;
3. the defect-completed square inverse $M_n^{-1} \binom{I}{0}$ and the rank-one projector identity;
4. the nodal ghost formula $D_n - I_X D_n = \omega_X + E_n - I_X E_n$;
5. the Chebyshev telescoping construction as a genuine convergent self-series in shifted copies of u ;
6. the universal endpoint-amplitude lower bound and its $\frac{1}{2}(\log 2)n^2$ consequence;
7. the finite q -binomial–Bell formula for geometric-node cardinal amplitudes.

These are described as new deductions within the pinned corpus boundary, not as unconditional claims of first publication in the world literature.

3 Rvachev’s up-function and the endpoint structure

3.1 Normalization and basic identities

Let F denote the Fabius function on $[0, 1]$ and let

$$u(x) = \text{up}(x) = F(1 - |x|), \quad -1 \leq x \leq 1,$$

extended by zero outside $[-1, 1]$ [1, 3, 17]. Then u is nonnegative, even, C^∞ , supported on $[-1, 1]$, satisfies $u(0) = 1$, and is flat at both endpoints. In the Fourier convention $\hat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx$,

$$\hat{u}(\xi) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{\xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}.$$

Equivalently, u is the density of

$$X = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad U_j \sim \text{Unif}[-1, 1]$$

with independent summands. Hence its moment generating function is

$$M_u(z) = \mathbb{E} e^{zX} = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}.$$

The odd cumulants vanish, while [17, 18]

$$\kappa_{2m} = \frac{B_{2m}}{2m(1 - 4^{-m})}. \tag{1}$$

This Bernoulli denominator is the source of the Bell-polynomial transform used below.

3.2 Flatness and the dyadic endpoint law

Set

$$g(t) = -\log F(2^{-t}).$$

A rigorous result in the primary exposition is [17]

$$\frac{g(t)}{t^2} \longrightarrow \frac{\log 2}{2} \quad (t \rightarrow \infty). \quad (2)$$

The sharper expansion is organized by the lower real branch W_{-1} of the Lambert function and contains a nonconstant periodic phase. The leading law (2) alone will already force a quadratic-exponential lower bound for every endpoint-basis realization of a nontrivial polynomial.

3.3 Why a broad atom can be tiny on the target interval

At degree n , the endpoint dictionary uses dilates of width 2^{n+1} whose centres lie far to the right of $[-1, 1]$. Their arguments on the target interval lie very near -1 , where $u(-1+s) = F(s)$ is extremely small. Exact polynomials therefore arise from enormous signed cancellations among very small positive functions. This geometric fact, rather than the choice of interpolation nodes, is the origin of the conditioning barrier proved in Section 14.

4 Lagrange interpolation on an arbitrary node set

Let

$$X_n = \{x_0, \dots, x_n\} \subset [-1, 1]$$

be distinct. It is convenient to map $[-1, 1]$ to $[0, 1]$ by

$$t = \frac{x+1}{2}, \quad \theta_j = \frac{x_j+1}{2}.$$

The Lagrange cardinal polynomials in the two coordinates are

$$\ell_{n,j}(x) = \prod_{i \neq j} \frac{x - x_i}{x_j - x_i}, \quad (3)$$

$$P_{n,j}(t) = \ell_{n,j}(2t-1) = w_j \prod_{i \neq j} (t - \theta_i), \quad w_j = \frac{1}{\prod_{i \neq j} (\theta_j - \theta_i)}. \quad (4)$$

The interpolation operator is

$$I_X f(x) = \sum_{j=0}^n f(x_j) \ell_{n,j}(x). \quad (5)$$

It reproduces Π_n , the polynomials of degree at most n .

The monic nodal polynomial is

$$\omega_X(t) = \prod_{j=0}^n (t - \theta_j). \quad (6)$$

Since $t^{n+1} - I_X(t^{n+1})$ is monic of degree $n+1$ and vanishes at every θ_j ,

$$t^{n+1} - I_X(t^{n+1}) = \omega_X(t). \quad (7)$$

This elementary identity becomes the polynomial part of the canonical ghost mode in Section 10.

Three node families will be distinguished.

Equispaced nodes

$x_j = -1 + 2j/n$. They give simple rational cardinals but exponentially large Lebesgue constants and Runge instability.

Chebyshev–Lobatto nodes

$x_j = \cos(j\pi/n)$. Their Lebesgue constants are $\mathcal{O}(\log n)$ and they are the stable choice for a global loop.

Geometric nodes

$\theta_j = q^j$, $0 < q < 1$, hence $x_j = 2q^j - 1$. Their cardinal weights admit closed q -binomial forms and connect the loop to the repository's q -Pochhammer calculus.

5 The imported endpoint polynomial-synthesis theorem

This section restates, in the notation needed here, the exact synthesis engine established in the repository [18]. Its main purpose is to isolate the single imported theorem on which the new cardinal and loop results depend.

5.1 The common-scale endpoint dictionary

Fix $n \geq 0$ and put

$$R_n = 2^n, \quad k_{n,r} = R_n - n - 1 + r, \quad 0 \leq r \leq n + 1.$$

On $0 \leq t \leq 1$, define

$$\psi_{n,r}(t) = \frac{1}{R_n} u\left(\frac{t - k_{n,r}}{R_n}\right). \quad (8)$$

The $n + 2$ functions $\psi_{n,0}, \dots, \psi_{n,n+1}$ form the right-endpoint basis of the local shift space.

5.2 The Bernoulli multiplier

Define

$$H_n(z) = \left(\frac{z}{2 \sinh(z/2)}\right)^n \frac{1}{M_u(z)}. \quad (9)$$

Using (1),

$$\log H_n(z) = - \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} \left(n + \frac{1}{1 - 4^{-m}}\right) z^{2m}. \quad (10)$$

Only terms through degree n matter when $H_n(D)$ acts on Π_n .

Let

$$A_n(z) = \prod_{m=1}^n [2^m]_z, \quad [N]_z = 1 + z + \dots + z^{N-1}, \quad A_n(z) = \sum_{s \geq 0} a_{n,s} z^s. \quad (11)$$

Then

$$A_n(1) = 2^{n(n+1)/2}. \quad (12)$$

Theorem 5.1 (Endpoint synthesis theorem; repository). *For every $P \in \Pi_n$ there is a unique vector $b(P) = (b_0, \dots, b_{n+1})^T$ such that*

$$P(t) = \sum_{r=0}^{n+1} b_r \psi_{n,r}(t) = \frac{1}{R_n} \sum_{r=0}^{n+1} b_r u\left(\frac{t - k_{n,r}}{R_n}\right), \quad 0 \leq t \leq 1. \quad (13)$$

Set

$$G = H_n(D)P, \quad g_r = A_n(1)G\left(r - \frac{n}{2}\right), \quad 0 \leq r \leq n+1.$$

Then b is the unique solution of the unit-triangular convolution

$$g_r = \sum_{s=0}^r a_{n,r-s} b_s, \quad 0 \leq r \leq n+1. \quad (14)$$

The arithmetic cost is $\mathcal{O}(n^2)$ once the coefficients of P are known.

Remark 5.2 (Why the scale 2^n is sharp). The Fourier transform of u has a dyadic zero-multiplicity pattern. At integer scale ratio ρ , the full translate family reproduces precisely through degree $\nu_2(\rho)$. Thus 2^n is the smallest integer ratio that can reproduce all of Π_n . Restriction to the active translates on one cell leaves an $(n+2)$ -dimensional local space: $n+1$ polynomial dimensions and one canonical defect dimension.

5.3 A self-contained algebraic inversion of the triangular step

Although [Theorem 5.1](#) is imported, its final inversion can be written in a useful closed form. Let

$$\frac{1}{A_n(z)} = \sum_{r \geq 0} \alpha_{n,r} z^r \quad (15)$$

as a formal power series. Since $a_{n,0} = 1$,

$$\alpha_{n,0} = 1, \quad \alpha_{n,r} = - \sum_{s=1}^r a_{n,s} \alpha_{n,r-s}. \quad (16)$$

Multiplying (14) by the inverse Toeplitz series gives

$$b_r = \sum_{s=0}^r \alpha_{n,r-s} g_s. \quad (17)$$

Thus the entire synthesis map is finite and explicit.

5.4 Bell-polynomial form of the differential multiplier

Write

$$\eta_{2m}^{(n)} = -\frac{B_{2m}}{2m} \left(n + \frac{1}{1-4^{-m}} \right), \quad \eta_{2m+1}^{(n)} = 0. \quad (18)$$

Then

$$H_n(z) = \exp \left(\sum_{q \geq 1} \eta_q^{(n)} \frac{z^q}{q!} \right) = \sum_{q \geq 0} Y_q \left(\eta_1^{(n)}, \dots, \eta_q^{(n)} \right) \frac{z^q}{q!},$$

where Y_q is the complete exponential Bell polynomial. Put

$$h_q^{(n)} = Y_q \left(\eta_1^{(n)}, \dots, \eta_q^{(n)} \right). \quad (19)$$

Then, for $P \in \Pi_n$,

$$H_n(D)P = \sum_{q=0}^n \frac{h_q^{(n)}}{q!} P^{(q)}. \quad (20)$$

All odd $h_q^{(n)}$ vanish.

6 Exact Lagrange polynomials as finite sums of shifted up-functions

We now apply the endpoint theorem to the individual cardinal polynomials. The result is the requested finite representation.

Theorem 6.1 (Exact common-dictionary cardinal synthesis). *Let $x_0, \dots, x_n$ be distinct points of $[-1, 1]$, and let $\ell_{n,j}$ be the corresponding Lagrange cardinal polynomials. Define*

$$c_{n,r} = 2^{n+1} - 2n - 3 + 2r, \quad 0 \leq r \leq n+1. \quad (21)$$

For every $j = 0, \dots, n$ there is a unique vector $b_j = (b_{j,0}, \dots, b_{j,n+1})^T$ such that

$$\ell_{n,j}(x) = \frac{1}{2^n} \sum_{r=0}^{n+1} b_{j,r} u\left(\frac{x - c_{n,r}}{2^{n+1}}\right), \quad -1 \leq x \leq 1. \quad (22)$$

All $n+1$ cardinals use the same $n+2$ atoms. If every x_j is rational, then every $b_{j,r}$ is rational.

Proof. Use $t = (x+1)/2$ and $\theta_j = (x_j+1)/2$. The polynomial $P_{n,j}(t) = \ell_{n,j}(2t-1)$ belongs to Π_n , so [Theorem 5.1](#) gives

$$P_{n,j}(t) = \frac{1}{2^n} \sum_{r=0}^{n+1} b_{j,r} u\left(\frac{t - (2^n - n - 1 + r)}{2^n}\right).$$

Substituting $t = (x+1)/2$ transforms the argument into

$$\frac{(x+1)/2 - (2^n - n - 1 + r)}{2^n} = \frac{x - (2^{n+1} - 2n - 3 + 2r)}{2^{n+1}},$$

which proves (22). Uniqueness follows from uniqueness in [Theorem 5.1](#). If the nodes are rational, then $P_{n,j} \in \mathbb{Q}[t]$; Bernoulli numbers, H_n , $A_n(1)$, the samples at half-integers, and the unit-triangular inverse are rational. Hence all $b_{j,r}$ are rational. $\square$

6.1 A closed finite coefficient formula

The theorem becomes fully explicit after combining Bell polynomials, elementary symmetric polynomials, and the inverse Toeplitz filter.

For a fixed j , let

$$\Theta^{(j)} = \{\theta_i : i \neq j\}.$$

Taylor expansion of the product in (4) gives

$$\frac{P_{n,j}^{(q)}(y)}{q!} = w_j e_{n-q}(\{y - \theta_i : i \neq j\}), \quad (23)$$

where e_m is the elementary symmetric polynomial and $e_0 = 1$. Consequently,

$$G_{n,j}(y) := H_n(D)P_{n,j}(y) = w_j \sum_{q=0}^n h_q^{(n)} e_{n-q}(\{y - \theta_i : i \neq j\}). \quad (24)$$

Only even q occur. Combining (17) with (24) proves the next result.

Corollary 6.2 (Bernoulli–Bell–symmetric formula). *The coefficients in (22) are*

$$b_{j,r} = A_n(1) \sum_{s=0}^r \alpha_{n,r-s} w_j \sum_{q=0}^n h_q^{(n)} e_{n-q} \left(\left\{ s - \frac{n}{2} - \theta_i : i \neq j \right\} \right). \quad (25)$$

All sums are finite; terms with odd q vanish.

Proof. By (17), $b_{j,r} = \sum_{s=0}^r \alpha_{n,r-s} g_{j,s}$, where $g_{j,s} = A_n(1) G_{n,j}(s - n/2)$. Substitute (24). $\square$

Remark 6.3. Formula (25) separates the three mathematical mechanisms: node geometry enters through w_j and the elementary symmetric functions; the probability law of u enters through Bernoulli/Bell coefficients $h_q^{(n)}$; and dyadic alias cancellation enters through A_n^{-1} and its Thue–Morse factorization.

6.2 Low-degree examples

Example 6.4 (Degree zero). The only cardinal is 1. With $R_0 = 1$,

$$1 = u(t) + u(t-1), \quad 0 \leq t \leq 1.$$

In x coordinates this is

$$1 = u\left(\frac{x+1}{2}\right) + u\left(\frac{x-1}{2}\right).$$

Example 6.5 (Degree one, nodes $-1, 1$). The common atoms have centres $-1, 1, 3$ and scale 4. Exact amplitudes are

$$\begin{aligned} \ell_{1,0}(x) &= \frac{1-x}{2} = \frac{3}{2}u\left(\frac{x+1}{4}\right) - u\left(\frac{x-1}{4}\right) + \frac{1}{2}u\left(\frac{x-3}{4}\right), \\ \ell_{1,1}(x) &= \frac{1+x}{2} = -\frac{1}{2}u\left(\frac{x+1}{4}\right) + u\left(\frac{x-1}{4}\right) + \frac{1}{2}u\left(\frac{x-3}{4}\right). \end{aligned}$$

Example 6.6 (Degree two, nodes $-1, 0, 1$). The common atoms are

$$U_r(x) = u\left(\frac{x - (1+2r)}{8}\right), \quad 0 \leq r \leq 3.$$

Writing $\ell_{2,j} = \sum_{r=0}^3 a_{j,r} U_r$, the exact raw amplitudes are

$$\begin{aligned} (a_{0,0}, a_{0,1}, a_{0,2}, a_{0,3}) &= \left(\frac{98}{9}, -\frac{188}{9}, \frac{170}{9}, -\frac{116}{9}\right), \\ (a_{1,0}, a_{1,1}, a_{1,2}, a_{1,3}) &= \left(-\frac{124}{9}, \frac{268}{9}, -\frac{268}{9}, \frac{124}{9}\right), \\ (a_{2,0}, a_{2,1}, a_{2,2}, a_{2,3}) &= \left(\frac{44}{9}, -\frac{98}{9}, \frac{116}{9}, -\frac{26}{9}\right). \end{aligned}$$

The large signed coefficients at such a low degree already foreshadow the conditioning results of [Section 14](#).

7 The geometric-node q -binomial specialization

Take

$$\theta_j = q^j, \quad 0 < q < 1, \quad 0 \leq j \leq n.$$

The denominator in the cardinal weight factors exactly:

$$\prod_{i \neq j} (q^j - q^i) = (-1)^j q^{jn-j(j+1)/2} (q; q)_j (q; q)_{n-j}, \quad (26)$$

where $(q; q)_m = \prod_{r=1}^m (1 - q^r)$. Hence

$$w_j = (-1)^j q^{\binom{j+1}{2} - jn} \frac{[n]_q}{(q; q)_n}. \quad (27)$$

The full nodal polynomial has the finite q -binomial expansion [17]

$$\prod_{i=0}^n (t - q^i) = t^{n+1} (t^{-1}; q)_{n+1} = \sum_{s=0}^{n+1} (-1)^s q^{\binom{s}{2}} \begin{bmatrix} n+1 \\ s \end{bmatrix}_q t^{n+1-s}. \quad (28)$$

For the elementary symmetric polynomial with q^j omitted, define $e_s^{(j)} = e_s(1, q, \dots, \widehat{q^j}, \dots, q^n)$. Since $e_s(1, q, \dots, q^n) = q^{\binom{s}{2}} [n+1]_q$, removing one variable gives

$$e_s^{(j)} = \sum_{r=0}^s (-1)^r q^{jr + \binom{s-r}{2}} \begin{bmatrix} n+1 \\ s-r \end{bmatrix}_q. \quad (29)$$

Proposition 7.1 (q -binomial–Bell cardinal formula). *For geometric nodes, every coefficient $b_{j,r}$ in (22) is a finite rational expression in q , Gaussian binomial coefficients, Bernoulli numbers, and complete Bell polynomials. Explicitly, insert (27) and the expansion*

$$P_{n,j}(t) = w_j \sum_{s=0}^n (-1)^s e_s^{(j)} t^{n-s} \quad (30)$$

into (20), sample at $r - n/2$, and apply (17).

Proof. Equations (26)–(30) are finite product identities. The operator $H_n(D)$ has coefficients $h_q^{(n)}$ given by Bernoulli numbers and Bell polynomials, and the final Toeplitz inverse uses the integer coefficients of $A_n(z)$. Every operation is a finite sum and product, proving the assertion. $\square$

Remark 7.2. Evaluating $P_{n,j}$ at $t = 0$ recovers the geometric Lagrange weights already used in the repository’s q -binomial identities. The present formula lifts those scalar evaluation weights to complete vectors of Rvachev synthesis amplitudes.

8 Substitution and exact closure of the loop

8.1 The double sum and its collapse

For any data values y_j , let

$$p_X(x) = \sum_{j=0}^n y_j \ell_{n,j}(x). \quad (31)$$

Substitute (22):

$$p_X(x) = \frac{1}{2^n} \sum_{j=0}^n y_j \sum_{r=0}^{n+1} b_{j,r} u \left(\frac{x - c_{n,r}}{2^{n+1}} \right)$$

$$= \frac{1}{2^n} \sum_{r=0}^{n+1} C_{n,r}(y) u\left(\frac{x - c_{n,r}}{2^{n+1}}\right), \quad (32)$$

where

$$C_{n,r}(y) = \sum_{j=0}^n y_j b_{j,r}. \quad (33)$$

Thus the nominal $(n+1)(n+2)$ atom occurrences collapse to $n+2$ distinct atoms.

Theorem 8.1 (Finite closed-loop identity). *Let f be any function defined at $x_0, \dots, x_n$. Then its Lagrange interpolant has the exact representation*

$$I_X f(x) = \frac{1}{2^n} \sum_{r=0}^{n+1} \left(\sum_{j=0}^n f(x_j) b_{j,r} \right) u\left(\frac{x - c_{n,r}}{2^{n+1}}\right) \quad (34)$$

for $-1 \leq x \leq 1$. In particular,

$$I_X u(x) = \frac{1}{2^n} \sum_{r=0}^{n+1} \left(\sum_{j=0}^n u(x_j) b_{j,r} \right) u\left(\frac{x - c_{n,r}}{2^{n+1}}\right). \quad (35)$$

Proof. Equation (34) is obtained by substituting the exact finite identity (22) into (5) and interchanging two finite sums. Taking $f = u$ gives (35). $\square$

8.2 Why direct synthesis gives the same answer

The polynomial p_X in (31) can be passed once through Theorem 5.1. Linearity of $H_n(D)$, of sampling, and of the triangular solve implies

$$b(p_X) = \sum_{j=0}^n y_j b(P_{n,j}). \quad (36)$$

Therefore the grouped coefficients in (33) are exactly the direct endpoint coefficients of the interpolating polynomial. The cardinal route is conceptually useful because it exposes the loop matrix; the direct route is computationally preferable when only one data vector is needed.

Corollary 8.2 (Complexity). *After construction of the interpolating polynomial in a stable basis, the direct loop coefficients can be computed in $\mathcal{O}(n^2)$ exact arithmetic operations. Precomputing the full cardinal synthesis matrix costs $\mathcal{O}(n^3)$ by applying the $\mathcal{O}(n^2)$ transform to $n+1$ cardinals, after which each new data vector costs $\mathcal{O}(n^2)$.*

9 The defect-completed matrix factorization

The loop has an especially transparent finite-dimensional form.

9.1 Sampling and synthesis matrices

Define the common atoms on $[-1, 1]$ by

$$\phi_{n,r}(x) = \frac{1}{2^n} u\left(\frac{x - c_{n,r}}{2^{n+1}}\right), \quad 0 \leq r \leq n+1. \quad (37)$$

Let

$$\Phi_X = (\phi_{n,r}(x_i))_{0 \leq i \leq n, 0 \leq r \leq n+1} \in \mathbb{R}^{(n+1) \times (n+2)} \quad (38)$$

be the sampling matrix, and let

$$B_X = (b_{j,r})_{0 \leq r \leq n+1, 0 \leq j \leq n} \in \mathbb{R}^{(n+2) \times (n+1)} \quad (39)$$

be the cardinal synthesis matrix, with cardinals indexed by columns. Evaluating (22) at the nodes gives

$$\Phi_X B_X = I_{n+1}. \quad (40)$$

Thus B_X is a right inverse of the rectangular sampling matrix.

9.2 The restricted Thue–Morse certificate

Let

$$\varepsilon_m = (-1)^{\text{wt}(m)}$$

be the signed Thue–Morse sequence. Restriction of the repository’s full polynomiality certificate to the endpoint block gives a row $\lambda_n^T \in \{\pm 1\}^{1 \times (n+2)}$ with entries

$$(\lambda_n)_r = (-1)^{r+1} \varepsilon_{\lfloor (2^{n+1} - n - 2 + r)/2 \rfloor}, \quad 0 \leq r \leq n+1. \quad (41)$$

For an endpoint coefficient vector b , the synthesized local function is a polynomial of degree at most n if and only if

$$\lambda_n^T b = 0. \quad (42)$$

Every cardinal column is polynomial, hence

$$\lambda_n^T B_X = 0. \quad (43)$$

Theorem 9.1 (Defect-completed inverse). *The square matrix*

$$M_X = \begin{pmatrix} \Phi_X \\ \lambda_n^T \end{pmatrix} \in \mathbb{R}^{(n+2) \times (n+2)} \quad (44)$$

is invertible, and

$$B_X = M_X^{-1} \begin{pmatrix} I_{n+1} \\ 0 \end{pmatrix}. \quad (45)$$

Proof. Suppose $M_X b = 0$. Then $\lambda_n^T b = 0$, so the synthesized local function $p_b = \sum_r b_r \phi_{n,r}$ belongs to Π_n . Also $\Phi_X b = 0$, so $p_b(x_j) = 0$ at $n+1$ distinct points. Hence $p_b = 0$. The endpoint atoms form a basis of the $(n+2)$ -dimensional local shift space, so $b = 0$. Therefore M_X is invertible. Equations (40) and (43) state precisely that $M_X B_X = \begin{pmatrix} I \\ 0 \end{pmatrix}$, yielding (45). $\square$

This formula is valuable computationally and conceptually. It says that the missing equation in the rectangular interpolation system is exactly the Thue–Morse defect constraint.

9.3 The coefficient projector

Define

$$\mathcal{P}_X = B_X \Phi_X \in \mathbb{R}^{(n+2) \times (n+2)}. \quad (46)$$

Then

$$\mathcal{P}_X^2 = B_X (\Phi_X B_X) \Phi_X = \mathcal{P}_X.$$

Moreover,

$$\text{ran}(\mathcal{P}_X) = \ker \lambda_n^T, \quad \ker \mathcal{P}_X = \ker \Phi_X. \quad (47)$$

Thus the loop is a projection from the full local shift space onto its polynomial hyperplane, along the unique local function vanishing at all interpolation nodes.

10 The canonical defect and the nodal ghost

10.1 Canonical defect normalization

The repository identifies a canonical generator of the one-dimensional nonpolynomial quotient of the local shift space:

$$D_n(t) = t^{n+1} + E_n(t), \quad (48)$$

where E_n is a smooth 1-periodic Fourier series supported on odd frequencies,

$$E_n(t) = -\frac{(n+1)!}{(2\pi i)^{n+1}} \sum_{m \in 2\mathbb{Z}+1} \frac{\hat{u}(m/2)}{m^{n+1}} e^{2\pi i m t}. \quad (49)$$

The defect coordinate of an endpoint coefficient vector b is

$$\delta_n(b) = \frac{\lambda_n^T b}{A_n(1)(n+1)!}. \quad (50)$$

The canonical D_n is normalized to have defect coordinate 1.

10.2 The ghost identity

Let I_Θ denote interpolation in the t variable at $\theta_0, \dots, \theta_n$.

Theorem 10.1 (Canonical nodal ghost). *The one-dimensional kernel of Φ_X is represented by*

$$\boxed{G_{n,X}(t) = D_n(t) - I_\Theta D_n(t) = \omega_X(t) + E_n(t) - I_\Theta E_n(t)}. \quad (51)$$

In particular, the ghost is the ordinary nodal polynomial corrected by the interpolation residual of the universal periodic defect.

Proof. Both D_n and $I_\Theta D_n \in \Pi_n$ belong to the local shift space, so their difference does also. It vanishes at every θ_j . It is nonzero because D_n has defect coordinate 1, whereas every polynomial has defect coordinate 0. Since the sampling map from an $(n+2)$ -dimensional space to $n+1$ nodal values has rank $n+1$ by (40), its kernel is one-dimensional; hence $G_{n,X}$ spans it.

Using (48), linearity of interpolation, and (7),

$$\begin{aligned} D_n - I_\Theta D_n &= (t^{n+1} - I_\Theta t^{n+1}) + (E_n - I_\Theta E_n) \\ &= \omega_X + E_n - I_\Theta E_n. \end{aligned}$$

□

10.3 Rank-one projector formula

Let v_X be the endpoint coefficient vector of $G_{n,X}$, normalized by

$$\lambda_n^T v_X = A_n(1)(n+1)!. \quad (52)$$

Corollary 10.2 (Exact rank-one decomposition). *The coefficient projector is*

$$\boxed{\mathcal{P}_X = I_{n+2} - \frac{v_X \lambda_n^T}{A_n(1)(n+1)!}}. \quad (53)$$

Equivalently, every endpoint coefficient vector b decomposes uniquely as

$$b = \mathcal{P}_X b + \frac{\lambda_n^T b}{A_n(1)(n+1)!} v_X, \quad (54)$$

where the first term is polynomial and the second is the nodal ghost.

Proof. By (47), $I - \mathcal{P}_X$ has range $\ker \Phi_X = \text{span}\{v_X\}$ and annihilates $\ker \lambda_n^T$. Therefore it has the form $v_X \mu^T$ for a row μ^T proportional to λ_n^T . Applying it to v_X and using (52) gives $\mu^T = \lambda_n^T / [A_n(1)(n+1)!]$. $\square$

Remark 10.3. Equation (53) is an algebraic closure of the loop: sampling and cardinal reconstruction remove exactly one coefficient-space mode, and that mode is the canonical defect adapted to the chosen nodes. Node design changes the polynomial correction $I_\Theta E_n$ but not the underlying periodic defect E_n .

11 Function-space operator interpretation

Let $S_n : \Pi_n \rightarrow C[-1, 1]$ denote endpoint synthesis in (13), transported to the x variable. By construction,

$$S_n p = p \quad \text{on } [-1, 1] \quad (55)$$

for every $p \in \Pi_n$. Define the closed-loop operator

$$T_{n,X} = S_n I_X. \quad (56)$$

Then

$$T_{n,X} f = I_X f \quad \text{on } [-1, 1]. \quad (57)$$

This simple identity has several consequences.

Proposition 11.1 (No hidden stabilization). *On the target interval, every approximation error, Lebesgue-constant bound, convergence theorem, or divergence phenomenon of the closed loop is exactly that of the original interpolation operator I_X .*

Proof. It is immediate from (57). $\square$

Outside $[-1, 1]$, $S_n I_X f$ supplies a compactly supported continuation by large up-atoms. That continuation may be useful for smooth windowing, but it cannot improve interpolation inside the interval.

12 Equispaced dyadic interpolation: exact finite closure but no justified series

The repository’s dyadic-comb volume studies global equispaced Lagrange interpolants of F and u [19]. It records exact barycentric weights, symmetry reductions, and rapid Runge growth. For example, the sampled error for the Fabius interpolant grows from approximately 4.99 at degree 32 to 3.59×10^7 at degree 64 and 3.84×10^{22} at degree 128. The precise values are evidence rather than a proof for every related node schedule, but they decisively warn against interpreting the finite cardinal sum as an automatically convergent series; compare the classical equispaced interpolation literature [11, 12].

Warning 12.1 (Finite identity versus infinite representation). For equispaced or dyadic global nodes, (35) is exact for every fixed n . It does *not* follow that the right-hand side converges to u as $n \rightarrow \infty$. Since it equals the original interpolant on $[-1, 1]$, any Runge divergence survives the substitution unchanged.

Our independent finite-convolution experiment shows the same qualitative behavior for symmetric u : the sampled equispaced error is already about 3.70×10^1 at degree 24, 4.50×10^2 at degree 32, and 2.76×10^6 at degree 48. These numbers depend on the numerical approximation and are not used as proof. They agree with the structural conclusion of Theorem 11.1.

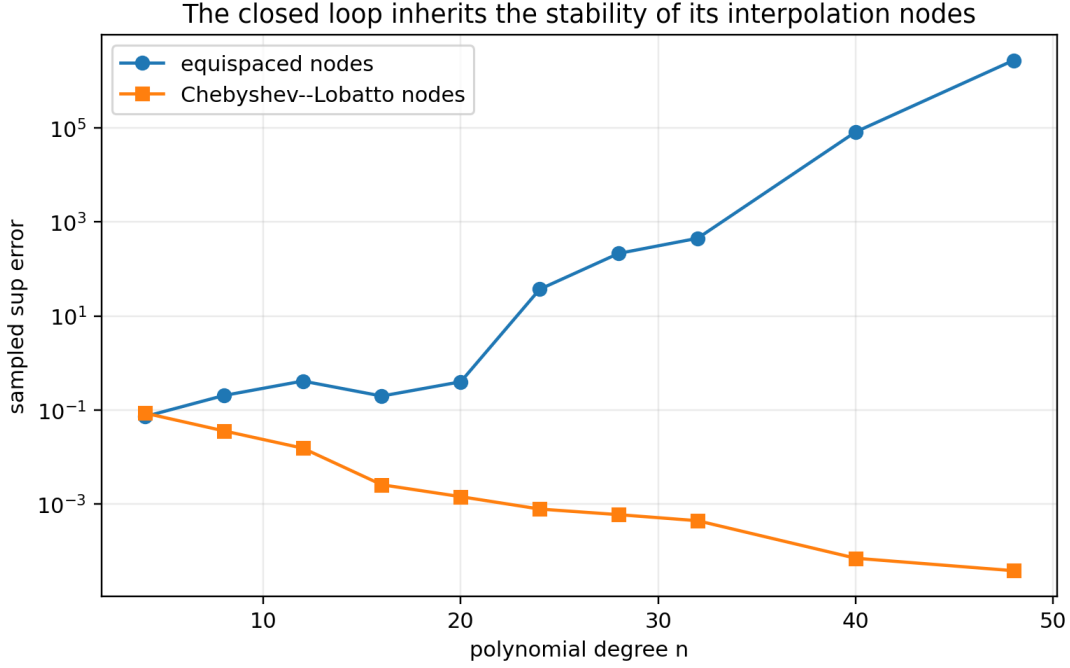


Figure 1: Sampled sup errors for a finite-convolution approximation of u . The closed loop has exactly these interpolation errors on $[-1, 1]$. Chebyshev–Lobatto nodes converge, whereas equispaced nodes exhibit Runge growth.

13 Chebyshev closure and a genuinely convergent self-series

13.1 Superalgebraic convergence

Let I_n^{CL} denote interpolation at the $n + 1$ Chebyshev–Lobatto nodes. Since $u \in C^\infty[-1, 1]$, its best polynomial approximation error satisfies

$$E_n(u) := \inf_{p \in \Pi_n} \|u - p\|_\infty = \mathcal{O}(n^{-m}) \quad \text{for every } m > 0. \quad (58)$$

The Chebyshev–Lobatto Lebesgue constant is $\mathcal{O}(\log n)$, so Lebesgue’s lemma gives [13, 14, 20]

$$\|u - I_n^{\text{CL}}u\|_\infty \leq (1 + \Lambda_n)E_n(u) = \mathcal{O}(n^{-p}) \quad \text{for every } p > 0. \quad (59)$$

Theorem 13.1 (Uniform superalgebraic closed loop). *Let $x_j = \cos(j\pi/n)$ and let $b_{j,r}^{\text{CL}}$ be the exact cardinal coefficients from Theorem 6.1. Then*

$$u(x) = \lim_{n \rightarrow \infty} \frac{1}{2^n} \sum_{r=0}^{n+1} \left(\sum_{j=0}^n u(x_j) b_{j,r}^{\text{CL}} \right) u\left(\frac{x - c_{n,r}}{2^{n+1}}\right) \quad (60)$$

uniformly for $x \in [-1, 1]$, and the function error is $\mathcal{O}(n^{-p})$ for every $p > 0$.

Proof. The finite sum in (60) equals $I_n^{\text{CL}}u$ by Theorem 8.1. The convergence and rate are therefore exactly (59). $\square$

13.2 Why the rate cannot be geometric

The Fabius function is a nowhere analytic C^∞ function, and so is the corresponding up-function on its interior pieces. If a fixed $\rho > 1$ and $C > 0$ satisfied

$$\|u - I_n^{\text{CL}} u\|_\infty \leq C\rho^{-n},$$

then the best approximation errors would obey the same bound. Bernstein’s converse theorem would imply analytic continuation to a Bernstein ellipse, contradicting nowhere analyticity.

Corollary 13.2 (Exact root rate). *For the Chebyshev closed loop,*

$$\limsup_{n \rightarrow \infty} \|u - I_n^{\text{CL}} u\|_\infty^{1/n} = 1. \quad (61)$$

Thus convergence is faster than every algebraic power but slower than every fixed geometric rate.

Proof. Superalgebraic convergence gives a limsup at most 1. If the limsup were strictly below 1, a geometric upper bound would hold for all sufficiently large n , contradicting the preceding argument. $\square$

13.3 A telescoping up-series

Set $N_m = 2^m$ and

$$J_m = I_{N_m}^{\text{CL}} u, \quad J_{-1} = 0, \quad Q_m = J_m - J_{m-1}.$$

Then

$$u = \sum_{m=0}^{\infty} Q_m \quad (62)$$

uniformly. In fact, for every $p > 0$,

$$\|Q_m\|_\infty \leq \|u - J_m\|_\infty + \|u - J_{m-1}\|_\infty = \mathcal{O}(2^{-pm}),$$

so the series is absolutely convergent in the Banach space $C[-1, 1]$. Synthesizing each $Q_m \in \Pi_{N_m}$ yields

$$Q_m(x) = \frac{1}{2^{N_m}} \sum_{r=0}^{N_m+1} d_{m,r} u\left(\frac{x - c_{N_m,r}}{2^{N_m+1}}\right). \quad (63)$$

Theorem 13.3 (Genuine convergent self-series). *The up-function admits the uniformly and absolutely convergent function-space series*

$$u(x) = \sum_{m=0}^{\infty} \frac{1}{2^{N_m}} \sum_{r=0}^{N_m+1} d_{m,r} u\left(\frac{x - c_{N_m,r}}{2^{N_m+1}}\right), \quad N_m = 2^m. \quad (64)$$

Absolute convergence here means $\sum_m \|Q_m\|_\infty < \infty$; it does not imply absolute convergence of the scalar atom amplitudes.

Proof. Equation (62) is a telescoping identity whose partial sums are J_m . The superalgebraic estimate proves absolute convergence in $C[-1, 1]$. Apply exact endpoint synthesis to each polynomial Q_m to obtain (63); substitution into (62) proves (64). $\square$

14 Endpoint conditioning and the Lambert-scale barrier

We now prove that the exact endpoint representation is intrinsically ill-conditioned in coefficient norm, irrespective of interpolation nodes.

14.1 A universal amplitude lower bound

Write the raw-amplitude form of (22) as

$$P(x) = \sum_{r=0}^{n+1} a_r u\left(\frac{x - c_{n,r}}{2^{n+1}}\right), \quad a_r = 2^{-n} b_r. \quad (65)$$

Let

$$\delta_n = \frac{n+2}{2^n}. \quad (66)$$

For every $x \in [-1, 1]$ and every endpoint centre $c_{n,r}$, a direct range calculation gives

$$-1 \leq \frac{x - c_{n,r}}{2^{n+1}} \leq -1 + \delta_n. \quad (67)$$

Since $u(-1 + s) = F(s)$ and F is increasing,

$$0 \leq u\left(\frac{x - c_{n,r}}{2^{n+1}}\right) \leq F(\delta_n). \quad (68)$$

Theorem 14.1 (Endpoint amplitude barrier). *Every representation (65) of a polynomial $P \in \Pi_n$ in the endpoint basis satisfies*

$$\boxed{\|a\|_1 \geq \frac{\|P\|_\infty}{F(\delta_n)}, \quad \|b\|_1 \geq \frac{2^n \|P\|_\infty}{F(\delta_n)}}. \quad (69)$$

In particular, for every Lagrange cardinal polynomial,

$$\|a_j\|_1 \geq \frac{1}{F(\delta_n)}. \quad (70)$$

Proof. From (68), for every $x \in [-1, 1]$,

$$|P(x)| \leq \sum_r |a_r| u\left(\frac{x - c_{n,r}}{2^{n+1}}\right) \leq F(\delta_n) \|a\|_1.$$

Take the supremum in x . Since $b = 2^n a$, the second inequality follows. A cardinal polynomial takes the value 1 at its own node, so its sup norm is at least 1. $\square$

14.2 Quadratic-exponential growth

Write

$$\delta_n = 2^{-t_n}, \quad t_n = n - \log_2(n+2). \quad (71)$$

Combining (70) with the dyadic law (2) yields the main asymptotic consequence.

Corollary 14.2 (Lambert-scale coefficient growth). *For every sequence of endpoint representations of normalized polynomials $P_n \in \Pi_n$ with $\|P_n\|_\infty \geq 1$,*

$$\log \|a^{(n)}\|_1 \geq \left(\frac{\log 2}{2} + o(1)\right) t_n^2, \quad (72)$$

and therefore

$$\boxed{\liminf_{n \rightarrow \infty} \frac{1}{n^2} \log \|a^{(n)}\|_1 \geq \frac{\log 2}{2}}. \quad (73)$$

The same leading lower bound holds for $\|b^{(n)}\|_1$.

Proof. Theorem 14.1 gives $\log \|a^{(n)}\|_1 \geq -\log F(2^{-t_n}) = g(t_n)$. Since $t_n \rightarrow \infty$, (2) proves (72). Finally $t_n/n \rightarrow 1$, giving (73). Multiplication by 2^n changes the logarithm by only $n \log 2 = o(n^2)$. $\square$

14.3 Stable output versus unstable internal coordinates

For Chebyshev nodes, the function-space interpolation operator has norm $\mathcal{O}(\log n)$ and converges superalgebraically on u . Nevertheless, every individual cardinal column of the endpoint synthesis matrix has ℓ^1 norm at least

$$\exp\left(\left(\frac{\log 2}{2} + o(1)\right)n^2\right).$$

Thus the map from nodal data to raw endpoint amplitudes is vastly more ill-conditioned than the map from nodal data to the represented function. This is exact cancellation, not approximation error.

A degree-4 numerical loop illustrates the phenomenon. The grouped raw amplitudes have ℓ^1 norm about 1.16×10^6 , although the represented polynomial has size about 1. The finite-convolution evaluation leaves a 2.3×10^{-4} discrepancy between the two visually coincident polynomial curves; the exact symbolic identity has zero discrepancy.

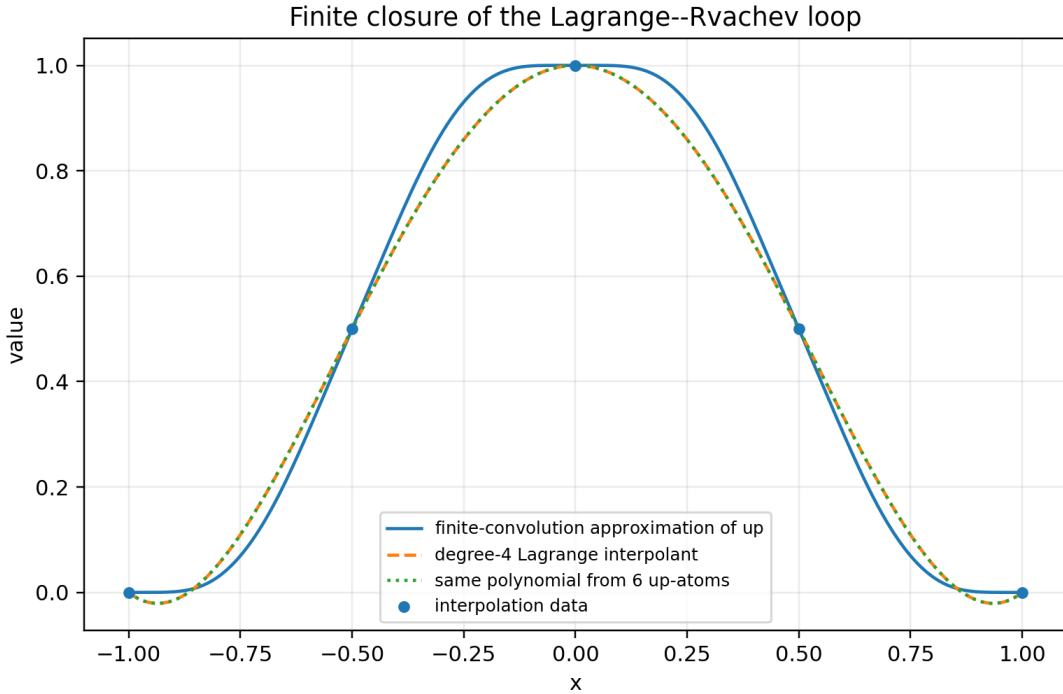


Figure 2: Finite degree-4 closure. The dashed Lagrange interpolant and dotted six-atom reconstruction are mathematically identical. Their small visible numerical discrepancy is caused by cancellation among amplitudes whose total variation is about 1.16×10^6 .

14.4 Exact coefficient data

The reproducibility script computes equispaced cardinal matrices exactly over $\mathbb{Q}$. Let

$$L_n = \max_{0 \leq j \leq n} \sum_{r=0}^{n+1} |2^{-n} b_{j,r}|.$$

Selected exact-arithmetic results are shown below.

n	$\log_{10} L_n$	endpoint-law proxy	L_n (scientific)
1	0.4771	0.0515	3.00×10^0
2	1.9401	0.0000	8.71×10^1
3	4.0171	0.0692	1.04×10^4
4	6.5522	0.3014	3.57×10^6
5	9.4620	0.7236	2.90×10^9
6	12.7880	1.3546	6.14×10^{12}
8	20.5134	3.2939	3.26×10^{20}
10	29.6674	6.1941	4.65×10^{29}
12	40.2033	10.1025	1.60×10^{40}
13	45.9670	12.4453	9.27×10^{45}

The “endpoint-law proxy” is $\frac{\log 2}{2 \log 10} t_n^2$ and records only the rigorous leading lower-bound scale. The large gap at moderate n reflects substantial lower-order terms, node conditioning, and non-sharpness of replacing all atom values by the single maximum $F(\delta_n)$.

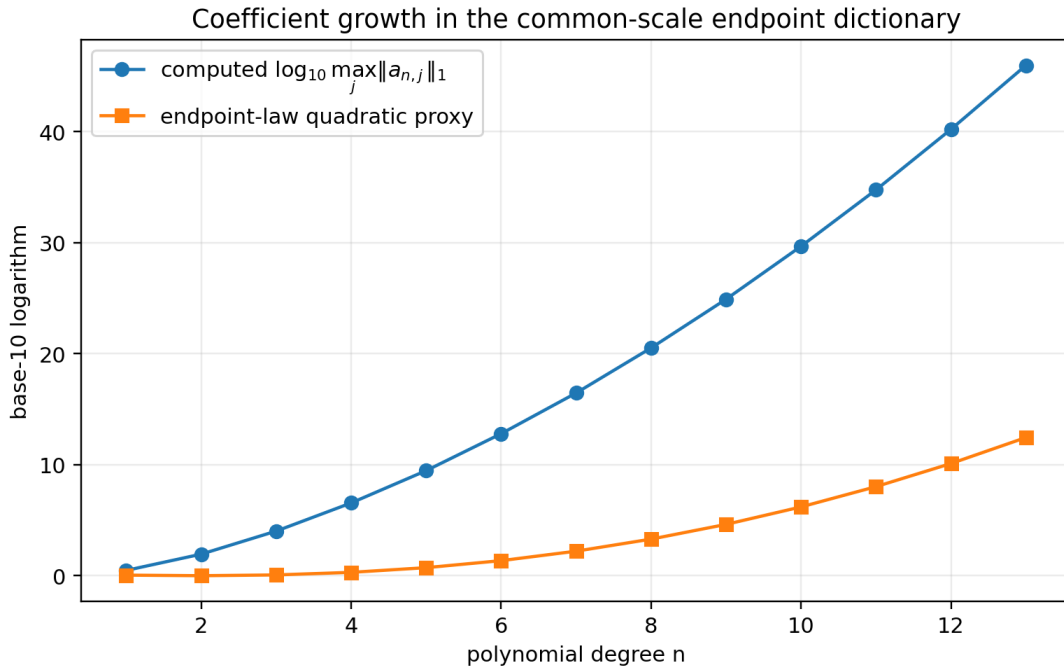


Figure 3: Exact rational coefficient norms through degree 13, compared with the leading quadratic endpoint-law proxy. Both curves are quadratic in scale; the computed cardinal coefficients contain large lower-order contributions.

15 Numerical experiments and reproducibility

15.1 Exact arithmetic component

The accompanying script `experiments.py` performs the following exact steps with SymPy rationals:

1. constructs the truncated logarithm (10) and exponentiates it through degree n ;
2. computes only the first $n + 2$ coefficients of $A_n(z)$, truncating every convolution;

3. builds each rational Lagrange cardinal polynomial;
4. applies $H_n(D)$, samples at $r - n/2$, and solves (14);
5. verifies the degree-0, degree-1, and degree-2 monomial examples;
6. verifies the restricted Thue–Morse certificate $\lambda_n^T B_X = 0$ through degree 13;
7. writes exact coefficient norms to `coefficient_growth.csv`.

15.2 Finite-convolution approximation of u

For plotting only, the script uses the probability representation $X = \sum 2^{-j} U_j$. After the first summand, the density is exactly constant on $[-1/2, 1/2]$. Convolution with the next uniform box of half-width a is implemented by

$$f_{\text{new}}(x) = \frac{F_{\text{old}}(x + a) - F_{\text{old}}(x - a)}{2a}, \quad (74)$$

where F_{old} is a numerically integrated CDF. Twenty boxes on a 2^{18} -interval grid give mass 1 and $u(0) \approx 0.99999237$.

15.3 Interpolation diagnostics

n	equispaced sampled error	Chebyshev–Lobatto sampled error
4	7.20×10^{-2}	8.54×10^{-2}
8	2.02×10^{-1}	3.58×10^{-2}
12	4.11×10^{-1}	1.53×10^{-2}
16	1.97×10^{-1}	2.55×10^{-3}
20	3.96×10^{-1}	1.42×10^{-3}
24	3.70×10^1	7.74×10^{-4}
28	2.14×10^2	5.88×10^{-4}
32	4.50×10^2	4.37×10^{-4}
40	8.24×10^4	6.95×10^{-5}
48	2.76×10^6	3.77×10^{-5}

The exact theory does not depend on these values. Their role is to show that the finite loop behaves exactly as the interpolation theorem predicts and to expose the numerical effect of coefficient cancellation.

16 Connections with the surrounding Fabius–Rvachev theory

16.1 Thue–Morse and the q -integer annihilator

The polynomial

$$A_n(z) = \prod_{m=1}^n [2^m]_z$$

is simultaneously a reproduction filter, a q -integer product, and the source of the Thue–Morse polynomiality certificate. The factorization

$$A_n(z)(z - 1)^{n+1} = (-1)^{n+1}(1 - z) \prod_{m=0}^n (1 - z^{2^m})$$

exhibits the finite Thue–Morse product. In the loop, this same object has three roles:

1. its low coefficients form the Toeplitz deconvolution in (14);
2. its value $A_n(1) = 2^{n(n+1)/2}$ sets the natural arithmetic scale;
3. its signed coefficient functional supplies the final row that makes the sampling matrix invertible.

Thus the Thue–Morse sequence does not enter only as a decorative identity; it is the exact algebraic separator between the polynomial hyperplane and the single local defect mode.

16.2 Bell and Bernoulli polynomials

The inverse moment transform $M_u(D)^{-1}$ and the scale correction $(D/[2 \sinh(D/2)])^n$ combine into $H_n(D)$. Their logarithms add, so cumulants linearize the synthesis problem. Complete Bell polynomials then convert cumulants to ordinary differential coefficients [8]. This is precisely why the cardinal formula (25) is finite: the degree- n polynomial kills every derivative above n .

The same mechanism can be expressed in the repository’s up-Appell family. If $\tilde{G}_m(y) = M_u(D)^{-1}y^m$, then $H_n(D)$ is obtained by composing that Appell inverse with the n th power of the centered-box deconvolution. The cardinal synthesis coefficients can therefore also be expanded in an Appell basis rather than monomials.

16.3 Infinite sinc products and Strang–Fix reproduction

The Fourier zeros of $\hat{u}(\xi) = \prod_j \text{sinc}(\xi/2^j)$ have dyadic multiplicities. At scale 2^n , nonzero lattice frequencies have enough zero multiplicity to reproduce all polynomials through degree n . This is the spectral reason the common scale is 2^n and the shift family has approximation order $n + 1$ [9, 10, 18]. The one extra local dimension is not a failure of Strang–Fix reproduction; it is the finite-cell alias defect isolated by E_n .

16.4 Lambert W and numerical conditioning

The lower branch W_{-1} describes the saddle scale at the flat endpoint of F [15, 17]. In the loop, the relevant endpoint argument is $\delta_n = (n + 2)2^{-n}$. Consequently, the exact large-amplitude cancellation is governed by the same Lambert phase that controls small-argument Fabius asymptotics. This is a direct bridge from asymptotic analysis to numerical linear algebra: the smallest atom magnitude bounds the inverse coordinate map.

16.5 Lagrange residual moments and the ghost

For arbitrary nodes, the residual of degree $n + 1$ is the nodal polynomial ω_X . For geometric nodes, higher residual moments are complete homogeneous symmetric polynomials and simplify to Gaussian-binomial expressions. The ghost identity

$$G_{n,X} = \omega_X + (I - I_X)E_n$$

shows how those classical interpolation residuals are modified by the canonical Rvachev defect. The polynomial and Fourier parts are therefore not separate phenomena; together they form the unique null mode of atom sampling.

16.6 Inverse Fabius and adaptive sampling

The endpoint coefficient barrier depends on $F(\delta_n)$. Any sharper asymptotic expansion for the inverse Fabius function immediately translates into a sharper statement about the degree at which a prescribed coefficient budget becomes impossible. Conversely, measured coefficient growth in carefully normalized dictionaries provides indirect numerical data about the endpoint scale. This duality suggests using inverse-Fabius asymptotics to choose adaptive degree and cell size in multi-cell synthesis.

17 Frontier conjectures and open problems

The following statements are plausible consequences of the exact framework and numerical evidence, but they are not proved here.

Conjecture 17.1 (Sharp Lambert–phase amplitude asymptotic). *For normalized endpoint cardinal columns in a regular node family,*

$$\log \|a_{n,j}\|_1 = -\log F((n+2)2^{-n}) + \Gamma_{X,j}(n) + o(1), \quad (75)$$

where $\Gamma_{X,j}(n)$ has at most order $n \log n$ and admits a decomposition into a smooth part plus a bounded nonconstant periodic function of the lower Lambert phase. After subtraction of the full known endpoint expansion, the normalized residual should be precompact and oscillatory rather than convergent.

Conjecture 17.2 (Universality of the quadratic constant). *Among all consecutive common-scale endpoint blocks at the sharp ratio 2^n , translation of the block can alter lower-order conditioning but cannot reduce the leading constant $\log 2/2$ in the logarithmic coefficient growth required to represent unit-size degree- n polynomials on one cell.*

Conjecture 17.3 (Central-basis improvement). *The unresolved central consecutive $(n+2)$ -atom basis is nonsingular for every n . After natural diagonal scaling by endpoint values and factorial/Appell weights, its condition number grows strictly more slowly than that of the right-endpoint basis, perhaps as $\exp(\mathcal{O}(n \log n))$ rather than $\exp(\Theta(n^2))$.*

Conjecture 17.4 (Ghost singular-vector principle). *For standard node families, the coefficient vector of $G_{n,X} = D_n - I_X D_n$ converges after diagonal normalization to the extremal right singular vector of the atom-sampling matrix. Its left singular partner is asymptotically the restricted Thue–Morse certificate.*

Conjecture 17.5 (Basic hypergeometric compression). *For geometric nodes $\theta_j = q^j$, the finite formula of Theorem 7.1 can be resummed into a terminating basic hypergeometric expression in j and r , with the Bell multiplier appearing as a finite q -difference deformation of the up-Appell family.*

Question 17.6 (Arithmetic valuations). *For dyadic rational nodes, how large are the odd denominator and 2-adic valuation of $b_{j,r}$? Real norms grow quadratically-exponentially, but the exact rational data suggest that the 2-adic structure may remain governed by short Thue–Morse recurrences. Can one obtain exact valuation formulas?*

Question 17.7 (Optimal total variation). *The endpoint basis is unique, so its amplitude norm is fixed. If arbitrary centres and scales are allowed, what is the minimum total variation*

$$\inf \left\{ \sum_s |a_s| : P(x) = \sum_s a_s u((x - c_s)/\sigma_s) \text{ on } [-1, 1] \right\}?$$

Can positivity, sign-change counts, or moment duality produce scale-independent lower bounds? Lagrange cardinals that change sign cannot admit a positive representation.

Question 17.8 (Multi-cell balance). *On a partition into J cells, the physical atom scale and atom count can be traded against each other. What choice of $J = J(n, \epsilon)$ minimizes total work under simultaneous approximation-error and coefficient-budget constraints? The answer should couple Chebyshev hp approximation with inverse-Fabius endpoint asymptotics.*

18 Algorithms

Algorithm 18.1 (Exact synthesis of one Lagrange cardinal). Input: degree n , distinct rational nodes $x_0, \dots, x_n$, and index j .

1. Set $\theta_i = (x_i + 1)/2$ and form $P_{n,j}(t) = \prod_{i \neq j} (t - \theta_i) / (\theta_j - \theta_i)$.
2. Construct $\log H_n(z)$ from (10), truncate at degree n , and exponentiate formally.
3. Compute $G = H_n(D)P_{n,j}$.
4. Form only $a_{n,0}, \dots, a_{n,n+1}$ by truncated convolution of the blocks $[2^m]_z$.
5. For $r = 0, \dots, n+1$, compute $g_r = A_n(1)G(r - n/2)$.
6. Solve $b_r = g_r - \sum_{s < r} a_{n,r-s}b_s$.
7. Return centres $c_{n,r}$ from (21) and amplitudes $2^{-n}b_r$.

Algorithm 18.2 (Closed loop for one data vector). Input: nodes x_j and data $y_j = f(x_j)$.

1. Construct the interpolating polynomial p_X in a numerically stable basis (barycentric, Newton, or Chebyshev).
2. Convert $p_X((2t-1))$ to a polynomial in t only when exact endpoint coefficients are required.
3. Apply the degree- n synthesis algorithm once to p_X .
4. Evaluate the $n+2$ grouped atoms. Use extended precision or compensated summation because the amplitude total variation can be enormous.

Algorithm 18.3 (Stable convergent loop). For approximation of u rather than symbolic identity:

1. use Chebyshev–Lobatto nodes, not a single global equispaced grid;
2. choose n from a Chebyshev coefficient tail or an a posteriori error estimate;
3. keep the polynomial representation for numerical evaluation whenever possible;
4. use the up-synthesis as an exact symbolic certificate, a compact smooth extension, or a multiscale building block;
5. if atom evaluation is unavoidable, use high precision predicted from $-\log F((n+2)2^{-n})$.

19 Formalization roadmap

The new results are well suited to incremental Lean formalization because the deep analytic ingredients are already represented in the repository.

1. Define the affine node transport and prove $P_{n,j}(t) = \ell_{n,j}(2t - 1)$.
2. Package the endpoint synthesis theorem as a linear equivalence between Π_n and the kernel of the certificate row.
3. Define B_X by applying this equivalence to the finite Lagrange basis and prove $\Phi_X B_X = I$.
4. Prove invertibility of M_X by the polynomial zero argument in [Theorem 9.1](#).
5. Formalize the rank-one projector using finite-dimensional linear algebra.
6. Introduce D_n and prove the ghost identity from $t^{n+1} - I_X t^{n+1} = \omega_X$.
7. Prove the endpoint amplitude bound from monotonicity of F and the explicit argument interval.
8. Combine with the already formalized or separately formalizable dyadic endpoint law to obtain the n^2 lower bound.

The only portion requiring substantial approximation-theory infrastructure is the Chebyshev convergence theorem. The finite algebraic loop and coefficient barrier are otherwise elementary consequences of existing exact objects.

20 Conclusions

The requested loop can be closed exactly, but the result has more structure than a formal substitution suggests.

First, every degree- n Lagrange cardinal on $[-1, 1]$ belongs to one common $n + 2$ -atom Rvachev dictionary. Its amplitudes are explicit combinations of node symmetric functions, Bernoulli numbers, complete Bell polynomials, and a q -integer Toeplitz inverse. The representation is finite and exact.

Second, substituting those formulas into a Lagrange interpolant does not create a large atom set: the double sum collapses to $n + 2$ atoms. At the matrix level, cardinal synthesis is a right inverse of atom sampling. The Thue–Morse defect row supplies the unique missing equation, and the resulting loop is a rank-one projection. Its null mode is the canonical defect minus its Lagrange interpolant, equivalently the nodal polynomial plus a universal periodic correction.

Third, finite closure and asymptotic convergence are different questions. Equispaced dyadic interpolation remains vulnerable to Runge divergence. Chebyshev–Lobatto interpolation yields a superalgebraically convergent loop and a genuine telescoping self-series, but no geometric rate is possible for the nowhere analytic up-function.

Finally, the endpoint basis has an unavoidable numerical price. Its atoms are evaluated at the flat edge of u , forcing raw amplitude norms of at least $\exp((\log 2/2 + o(1))n^2)$. This converts the Lambert- W endpoint asymptotics of the Fabius function into a sharp qualitative statement about coordinate conditioning. The most promising next step is therefore not merely to push degree higher, but to optimize the dictionary phase, scale, and cell partition while preserving the exact Thue–Morse/Appell algebra.

A Detailed proof of the elementary-symmetric derivative identity

Let

$$P(t) = w \prod_{i=1}^n (t - \beta_i).$$

For an auxiliary variable h ,

$$P(y + h) = w \prod_{i=1}^n ((y - \beta_i) + h) = w \sum_{q=0}^n e_{n-q}(y - \beta_1, \dots, y - \beta_n) h^q.$$

Taylor’s formula also gives

$$P(y + h) = \sum_{q=0}^n \frac{P^{(q)}(y)}{q!} h^q.$$

Equating coefficients proves (23). Substitution into (20) proves (24) without any appeal to a symbolic algebra system.

B Proof of the geometric excluded-variable identity

Let $E_s = e_s(1, q, \dots, q^n)$ and $E_s^{(j)} = e_s(1, q, \dots, \hat{q}^j, \dots, q^n)$. Splitting terms by whether they use q^j gives

$$E_s = E_s^{(j)} + q^j E_{s-1}^{(j)}.$$

Iterating,

$$E_s^{(j)} = E_s - q^j E_{s-1} + q^{2j} E_{s-2} - \dots + (-1)^s q^{sj} E_0.$$

The finite q -binomial theorem gives $E_r = q^{\binom{r}{2}} \begin{bmatrix} n+1 \\ r \end{bmatrix}_q$, proving (29).

C Exact low-degree regression identities

The implementation checks the following imported monomial syntheses in the t coordinate:

$$\begin{aligned} 1 &= u(t) + u(t-1), \\ t &= -\frac{1}{2}u\left(\frac{t}{2}\right) + u\left(\frac{t-1}{2}\right) + \frac{1}{2}u\left(\frac{t-2}{2}\right), \\ t^2 &= \frac{13}{9}u\left(\frac{t-1}{4}\right) - \frac{31}{9}u\left(\frac{t-2}{4}\right) + \frac{49}{9}u\left(\frac{t-3}{4}\right) + \frac{5}{9}u\left(\frac{t-4}{4}\right). \end{aligned}$$

These identities are used only as regression tests; the general algorithm is not inferred from them.

D Reproducibility files

The archive accompanying this report contains:

Lagrange_Rvachev_Loop.tex

The complete LaTeX source.

Lagrange_Rvachev_Loop.pdf

The compiled report.

experiments.py

Commented exact and numerical code.

coefficient_growth.csv

Exact-arithmetic coefficient norm data through degree 13.

numerical_summary.txt

Plain-text numerical diagnostics.

coefficient_growth.png, interpolation_stability.png, finite_loop_demo.png

Figures used in the report.

E Repository paths audited most directly

The repository root is <https://github.com/VladimirReshetnikov/ProveIt>. The most directly used source paths, written relative to that root, are:

- Analysis/FabiusFunction/docs/
- Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
- Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/Up_Polynomial_Synthesis/Up_Polynomial_Synthesis.tex
- Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/inverse-and-sampling/Dyadic_Comb_Frontiers/Dyadic_Comb_Frontiers.tex
- Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/Representation_Frontiers/Representation_Frontiers.tex

References

- [1] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.
- [2] V. L. Rvachev and V. A. Rvachev, A certain finite function, *Dopovidi Akademii Nauk Ukrain's'koi RSR, Series A* (1971), 705–707, 764.
- [3] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.
- [4] J. Arias de Reyna, Definición y estudio de una función indefinidamente diferenciable de soporte compacto, *Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid* **76** (1982), no. 1, 21–38; English translation, arXiv:1702.05442.
- [5] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Article A51; arXiv:1702.06487, version 3.
- [6] J. K. Haugland, Evaluating the Fabius function, arXiv:1609.07999 (2016).
- [7] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.

- [8] L. Comtet, *Advanced Combinatorics*, Reidel, 1974.
- [9] G. Strang and G. Fix, A Fourier analysis of the finite element variational method, in *Constructive Aspects of Functional Analysis*, C.I.M.E., 1973, 793–840.
- [10] C. de Boor and A. Ron, Fourier analysis of the approximation power of principal shift-invariant spaces, *Constructive Approximation* **8** (1992), 427–462.
- [11] C. Runge, Über empirische Funktionen und die Interpolation zwischen äquidistanten Ordinaten, *Zeitschrift für Mathematik und Physik* **46** (1901), 224–243.
- [12] T. M. Mills and S. J. Smith, The Lebesgue constant for Lagrange interpolation on equidistant nodes, *Numerische Mathematik* **61** (1992), 111–115.
- [13] J.-P. Berrut and L. N. Trefethen, Barycentric Lagrange interpolation, *SIAM Review* **46** (2004), no. 3, 501–517.
- [14] L. N. Trefethen, *Approximation Theory and Approximation Practice*, SIAM, 2013.
- [15] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359.
- [16] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/>.
- [17] V. Reshetnikov, *Fabius Function and Rvachev Up: Primary Exposition*, ProveIt repository, current source inspected 29 August 2026.
- [18] V. Reshetnikov, *Exact Polynomial Synthesis by Finite Rvachev Up-Function Dictionaries* (consolidated as *Up Polynomial Synthesis*), ProveIt repository, current source inspected 29 August 2026.
- [19] V. Reshetnikov, *Dyadic Comb Frontiers*, ProveIt repository, current source inspected 29 August 2026.
- [20] V. Reshetnikov, *Representation Frontiers*, ProveIt repository, current source inspected 29 August 2026.